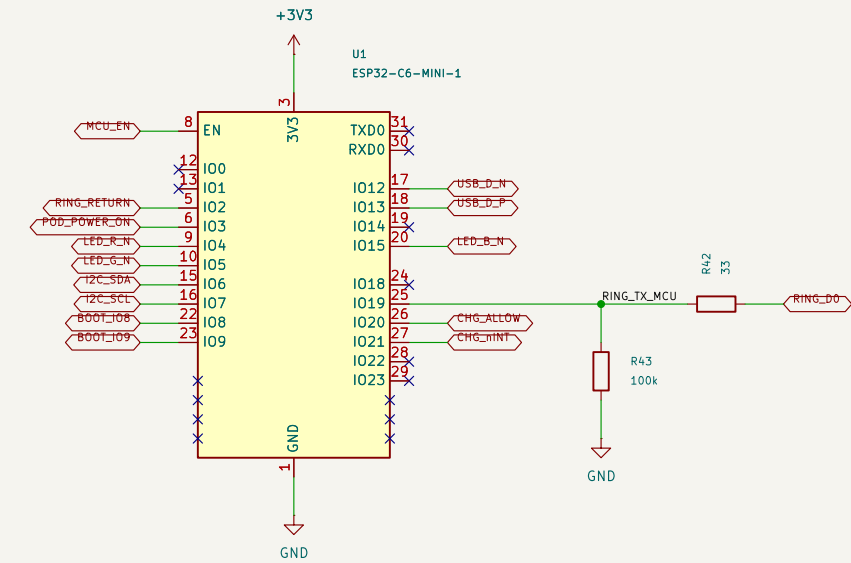


HAPTIC BRACELET / MAIN PCBA

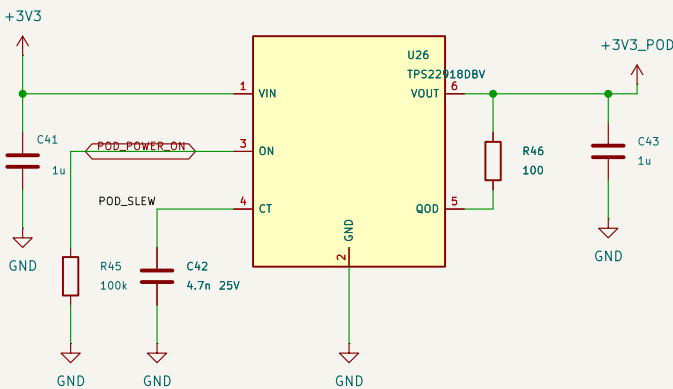
250 kbaud logical ring: ESP TX -> pod 0 -> ... -> pod 7 -> RETURN -> ESP RX.
Physical five-wire chain; RETURN runs back through the same links. No electrical connection across clasp.



TX pulldown keeps the ring quiet while U1 resets.
Before pod power OFF: detach UART, TX low;
RX input with pull-up disabled. No signal isolators.

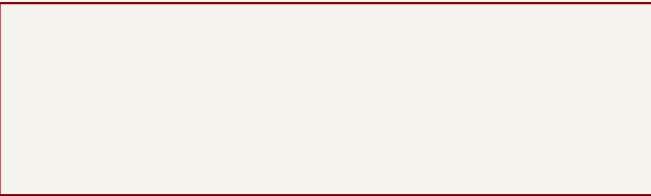
GPIO19 ring TX / GPIO2 ring RX
GPIO18 spare (no connect)
GPIO3 high powers pods
GPIO6/7: charger I2C only
GPIO0/1/14/22/23 spare (USB controls removed)
UART0 pads removed; native USB handles programming/debug

Pod recovery: cycle +3V3_POD through U26.
No shared reset wire; local ICE reset only.



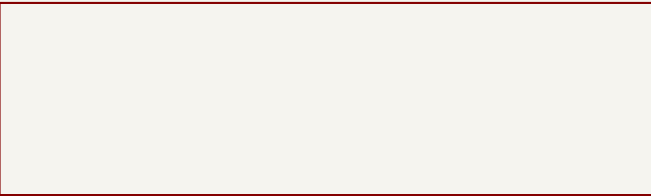
POD SUPPLY: default OFF; ESP, USB and charger stay powered.
CT controls ramp; Q0D discharges the ring after shutdown.
Do not enable UART idle-high until POD_3V3 has settled.
Disable charging before removing required pod temperature sensing.

Local haptic pod and harness



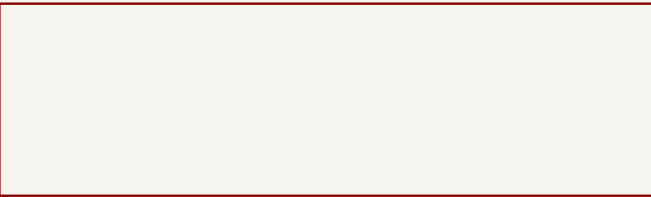
File: local-pod.kicad_sch

Charger and regulated supply



File: power.kicad_sch

USB-C data and power



File: usb.kicad_sch

ESP reset, boot and decoupling



File: controller-support.kicad_sch

One wrist / ring pod schematic draft

Sheet: /
File: main.kicad_sch

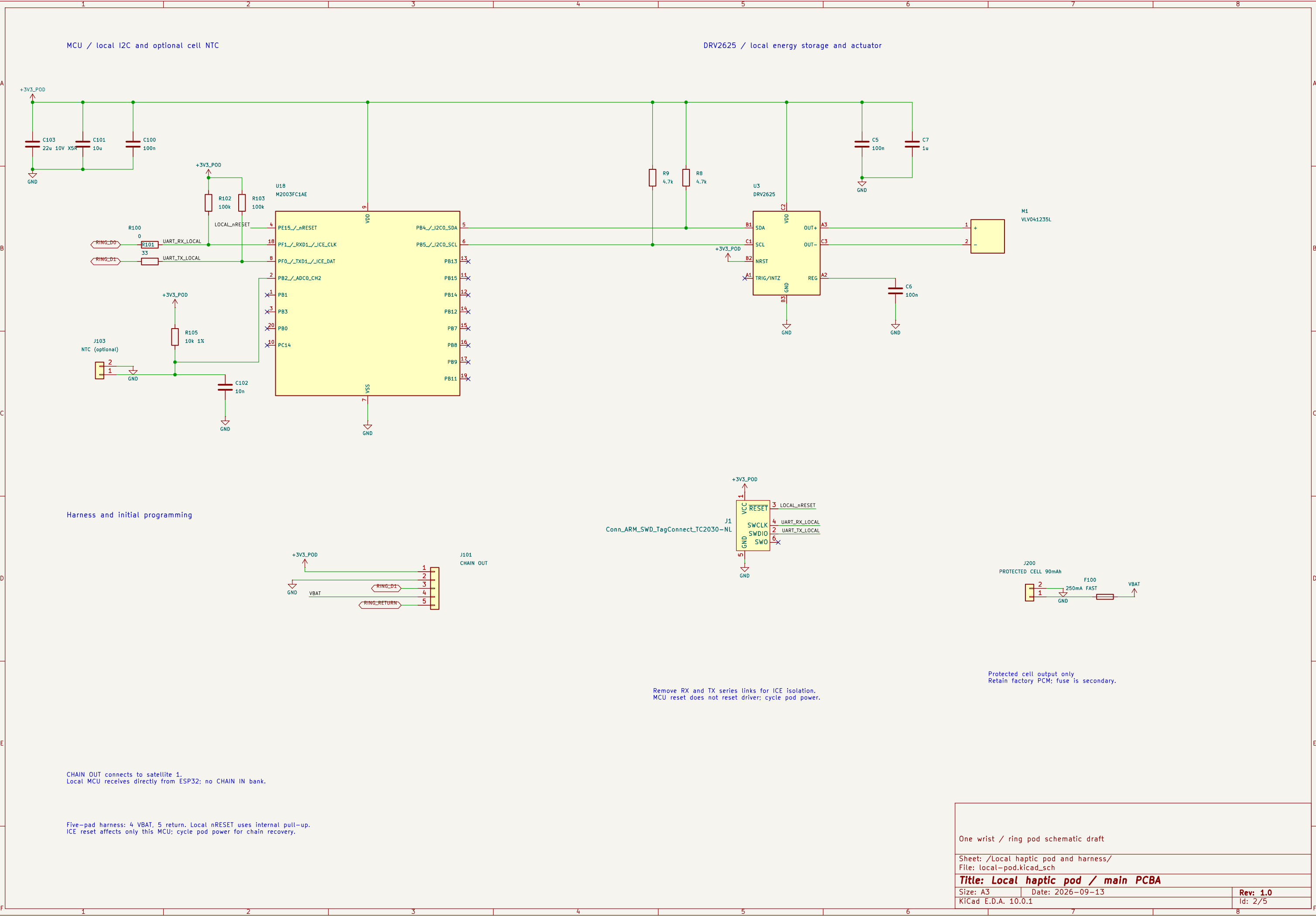
Title: Controller and switched pod power

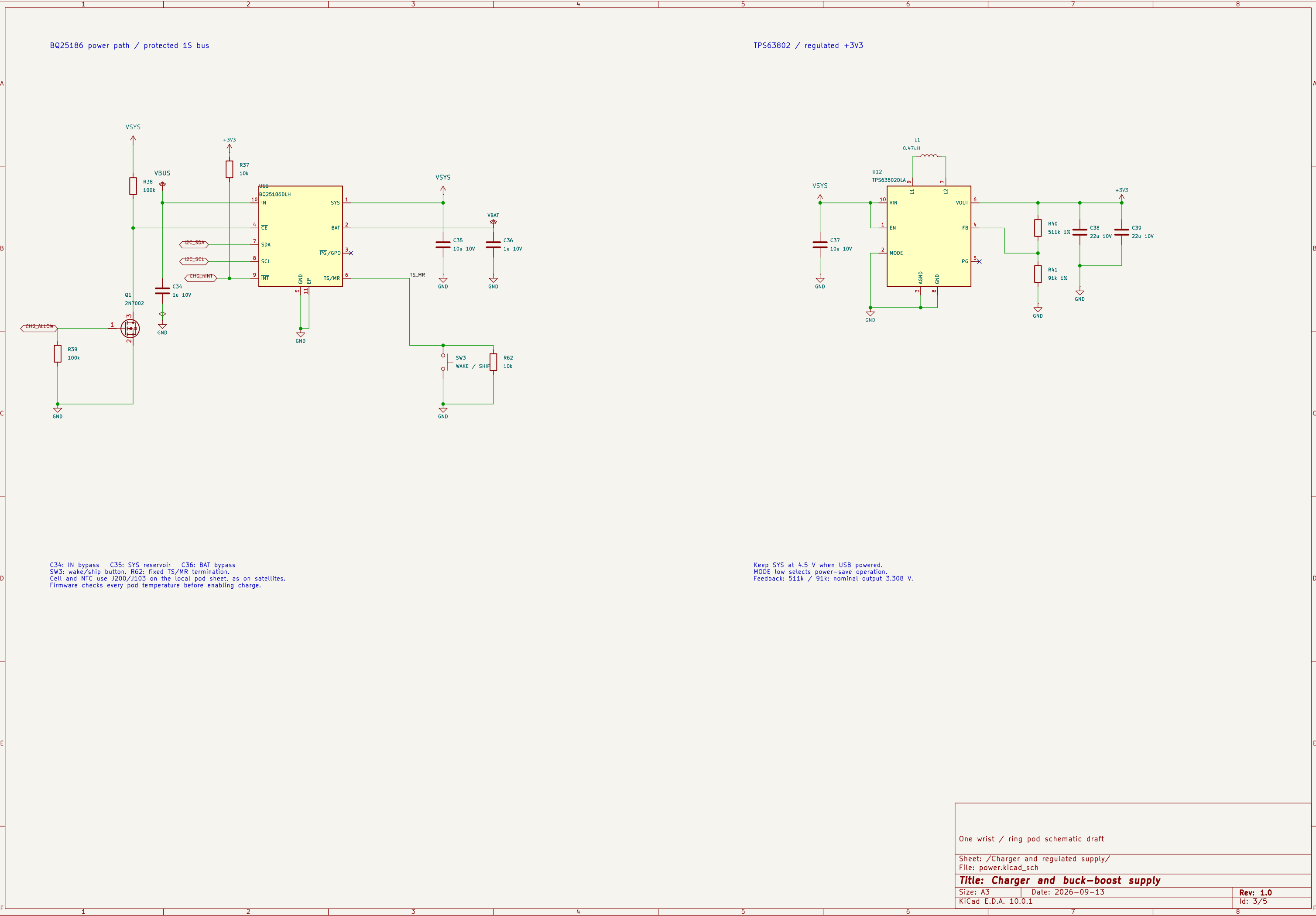
Size: A3 Date: 2026-09-12

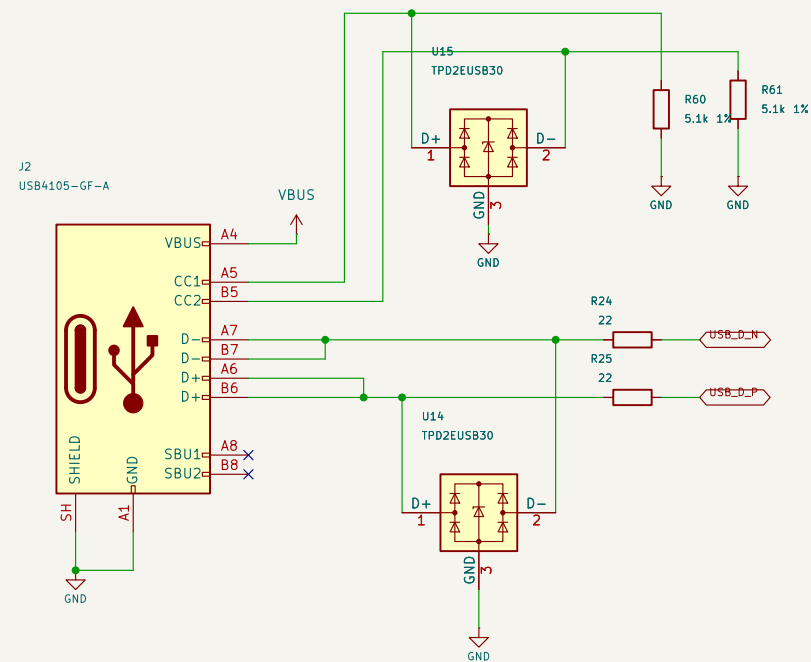
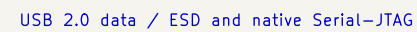
KiCad E.D.A. 10.0.1

Rev: 0.5

Id: 1/5







R60 and R61 each terminate one CC pin to GND.
USB 2.0 contacts are paired for either plug orientation.
C29/C30 remain optional DNP tuning capacitors.

Connector bypass



USB_VBUS connects directly to charger U11 IN and C34.
BQ25186 provides input-current limiting; no external current selector.
No CC current-advertisement detection or USB PD negotiation.
Read VIN_PGOOD_STAT over charger I2C for usable input-power status.
No discrete VBUS detector or UART service bank.

One wrist / initial circuit

Sheet: /USB-C data and power/
File: usb.kicad_sch

Title: USB-C / passive CC, native USB and VBUS sensing

Size: A3	Date: 2026-09-13	Rev: 1.0
KiCad E.D.A. 10.0.1		Id: 4/5

